Solution 1: Map the Roman Empire

This solution follows the steps of the exercise. Other choices (colours, sizes, classification modes) are fine, as long as the maps show what the exercise asks.

A) Map the empire and its landscape

1. **Load the data.** Drag `roman_empire.gpkg` into QGIS and select the three layers (`provinces`, `cities`, `monuments`). Then drag in `roads.gpkg` and `elevation.tif`. See Importing vector and raster data.
2. **Layer order.** In the *Layers* panel, drag the layers into this order, from top to bottom: `monuments`, `cities` (points), `roads` (lines), `provinces` (polygons), `elevation` (raster). The layer at the top is drawn on top. You can untick `monuments` for now: you do not need it for Map 1.
3. **Elevation.** Open the *Layer Properties* of `elevation` (double-click the layer) and go to *Symbology*.
   - Set *Render type* to *Singleband pseudocolor*.
   - Choose a *Color ramp* that runs from green to red or brown, for example *RdYlGn*. This ramp runs from red to green, so choose *Invert Color Ramp* in the drop-down list of the ramp: low land then becomes green and mountains red.
   - Click *Apply*. The sea is NoData and is not drawn.
   - With the *Identify Features* tool  , click a city in the Alps with `elevation` selected in the *Layers* panel, for example *Augusta Praetoria* (Aosta). QGIS shows the value of the cell. The cells are about 4.6 km wide, so the value is an average for the area around the city, not the exact elevation of the city itself.
4. **Provinces as outlines.** In the *Symbology* of `provinces`, click *Simple Fill* and set *Fill style* to *No Brush*. Choose a dark grey *Stroke color*. The elevation now stays visible under the provinces.
5. **Roads.** In the *Symbology* of `roads`, choose *Categorized*, set *Value* to `major` and click *Classify*. Double-click the symbol of `yes` and give it a larger *Width* (for example 0.6 mm) than the symbol of `no` (for example 0.2 mm). You can delete the category *all other values*: every road has `yes` or `no`.
6. **Cities.** In the *Symbology* of `cities`, choose *Categorized* on `type` and click *Classify*. Give `city` a small dot (*Simple Marker*, size 1.2 mm) and `legionary fortress` a square (choose the square shape in the *Simple Marker* settings, size 2.4 mm).
   - Go to *Labels*, choose *Single Labels* and set *Value* to `name`.
   - At the scale of the whole empire, there are too many labels to read. QGIS then leaves out labels that would overlap. Zoom in on a region to see more of them. You can also show the labels only from a certain scale (*Rendering* ▸ *Scale dependent visibility* in the *Labels* tab): see Style versus scale.
7. **Save Map 1 as a map theme.** Click the eye icon at the top of the *Layers* panel (*Manage Map Themes*) and choose *Add Theme…*. Call the theme "Map 1" and click *OK*. See Saving and reusing styles.

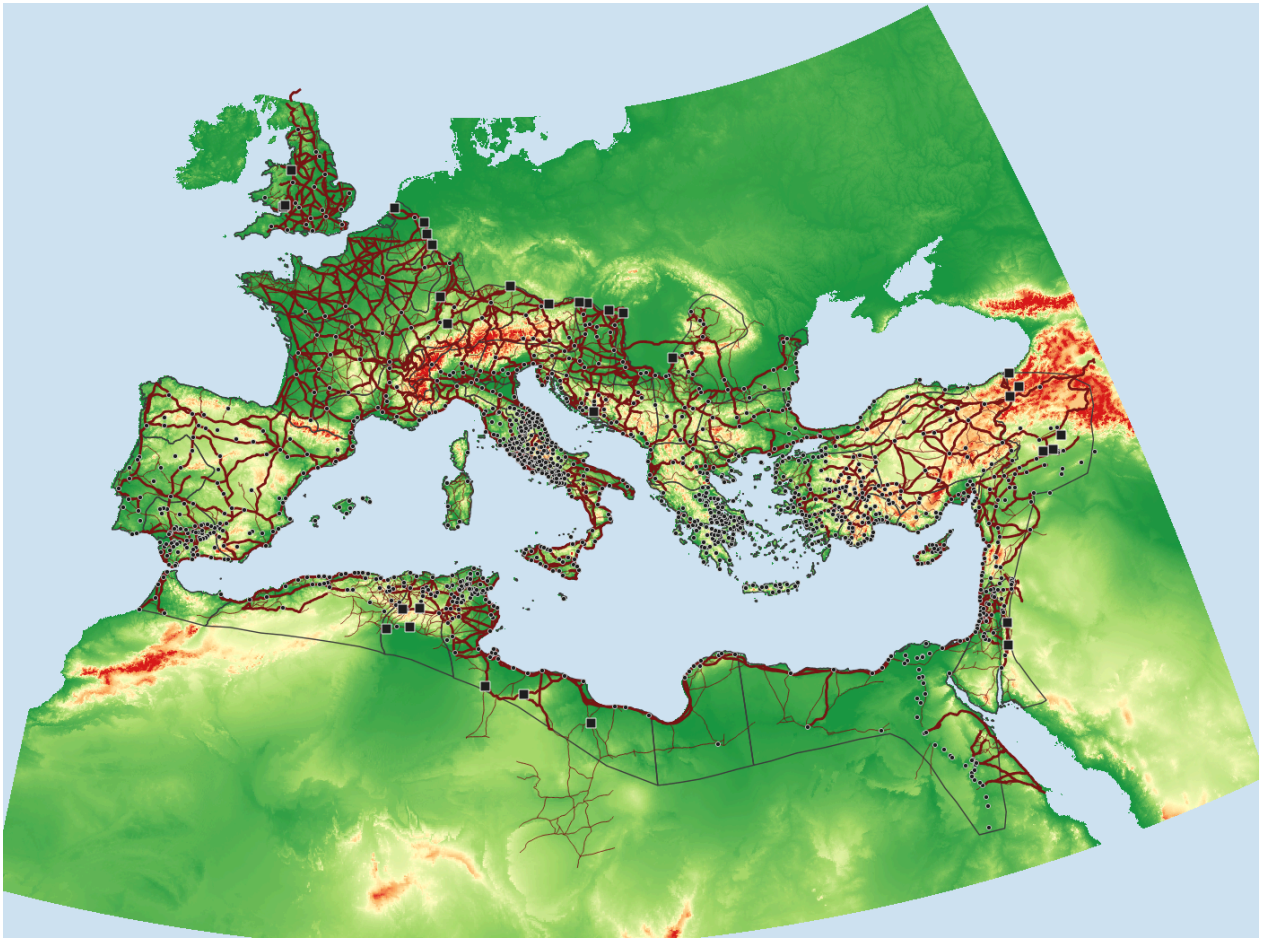


Figure 1: Map 1: elevation, province borders, roads (major roads thicker) and cities (squares = legionary fortresses), shown in EPSG:3035 (step C).

B) Map the infrastructure per province

You make this map in the same project: untick `elevation`, `roads` and `cities`.

1. **A second style for the provinces.** Right-click `provinces` and choose *Styles* ▸ *Add…*. Call the new style "Map 2" and click *OK*. The new style starts as a copy of the current one (outlines only) and is now the active style: every change you make from now on is stored in "Map 2", and the style of Map 1 stays as it is.

2. **Graduated provinces.** In the *Symbology* of `provinces`, choose *Graduated* and set *Value* to `area_km2`.

   - Choose a *Color ramp* in one colour, from light to dark (for example *Greens*).
   - Choose a *Mode*, for example *Natural Breaks (Jenks)*, with 5 *Classes*, and click *Classify*. See Graduated symbology.
   - Go to *Labels*, choose *Single Labels* and set *Value* to `name`.

3. **Pie charts.** Go to the *Diagrams* tab of `provinces` and choose *Pie Chart*. See Diagrams.

   - In *Attributes*, add the fields `n_theatres`, `n_amphitheatres`, `n_circuses`, `n_baths` and `n_temples` (double-click each field, or select it and click the plus button). Give each field a clear colour and a legend name (theatres, amphitheatres, circuses, baths, temples).
   - In *Size*, choose *Scaled size* and set *Attribute* to `n_monuments`, the total of the five fields. Click *Find* to fill in the *Maximum value*, and set the *Size* of the largest pie (for example 15 mm).
   - In *Placement*, keep *Over Centroid*.

- Click *OK*.

On a map with many pie charts, QGIS leaves out labels that would overlap with the pies. That is why you may see only a few province names.

4. **Save Map 2.** Untick `elevation`, `roads` and `cities` (if you have not done so yet), then eye icon ▸ *Add Theme…*, called "Map 2". Now choose eye icon ▸ *Map 1*: QGIS shows the elevation, roads and cities again, and the provinces as outlines. Eye icon ▸ *Map 2* brings back the graduated provinces with the pie charts.

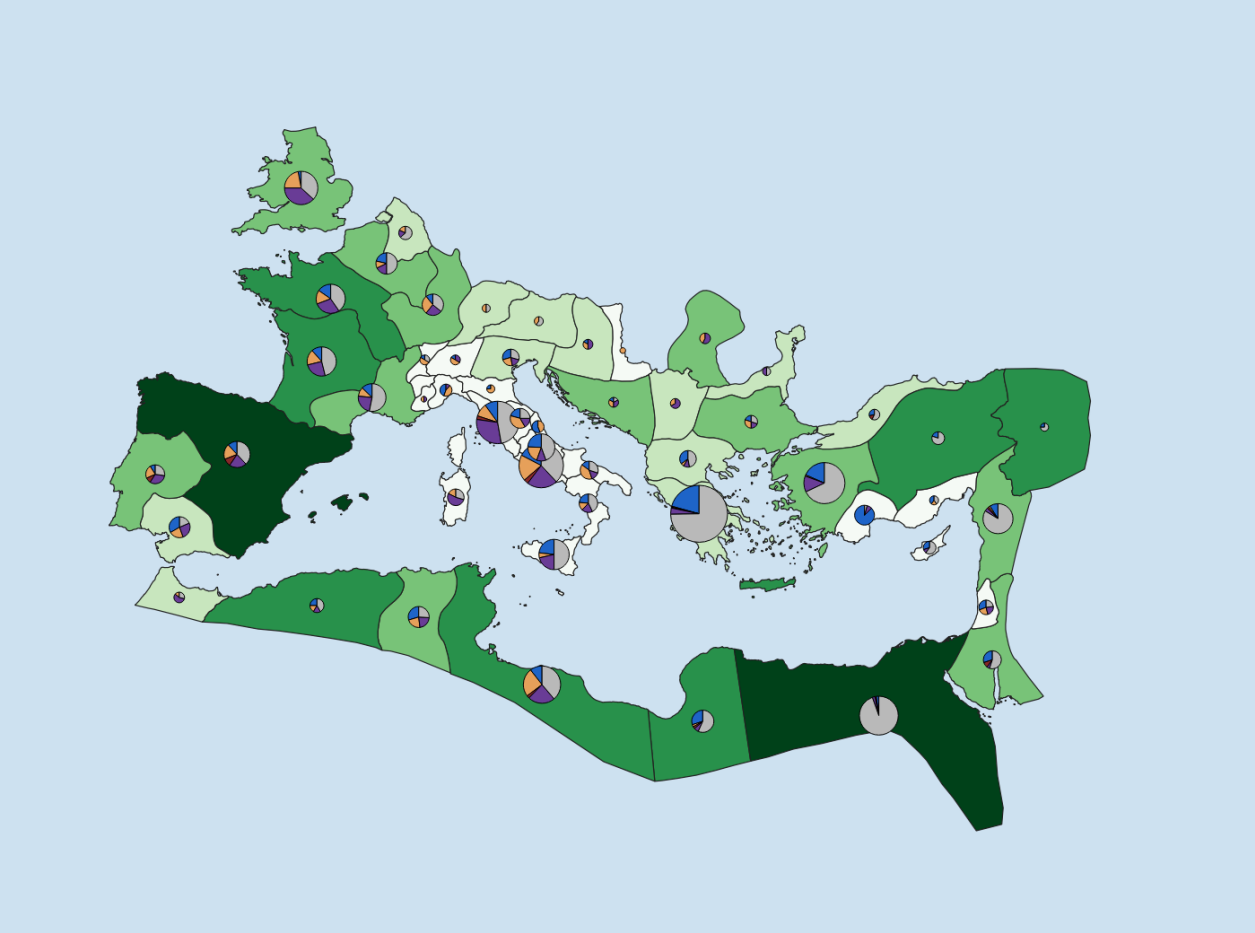


Figure 2: Map 2: provinces coloured by area, with pie charts of the monuments per type (theatres blue, amphitheatres orange, circuses red, baths purple, temples grey); the size shows the total number. Shown in EPSG:3035 (step C).

C) Prepare for finishing

1. **Check the CRS.** Right-click each layer, choose *Properties…* and look at the *Source* tab: see Coordinate system of the data. The CRS of the project is shown at the right end of the status bar.
   - `provinces`, `cities`, `monuments` and `elevation` are in EPSG:4326 (WGS 84), `roads` is in EPSG:3035.
   - The project is in EPSG:4326 if you loaded `roman_empire.gpkg` first: by default, QGIS gives a new project the CRS of the first layer you add.
   - So the layers do not all have the same CRS, and `roads` does not match the project. For this map, that does not matter: QGIS transforms `roads` on the fly, so all layers line up. If you want to analyse layers together, it is better to transform them to one CRS first: see Why transform data explicitly?.

2. **A suitable coordinate system.** Choose *Project* ▸ *Properties…*, go to *CRS*, type `3035` in the *Filter* box, select *ETRS89-extended / LAEA Europe* and click *OK*. QGIS now transforms all layers on the fly. The provinces in the north (Britannia, Germania) are no longer stretched in width, as they are in EPSG:4326, and the empire looks as it does on a globe. EPSG:3035 is made for Europe: the provinces furthest away (Aegyptus, Mesopotamia) are a little more distorted, but much less than in EPSG:4326. It is also an equal-area projection, which suits Map 2, where the colours show the area of the provinces. See Coordinate system of the QGIS project. If QGIS asks which transformation to use, keep the proposed one.

3. **Save the project.** Choose *Project* ▸ *Save As…* and save the project in the folder of the dataset, for example as `roman_empire.qgz`. The project only stores where the data are and how they look, not the data themselves: see The QGIS project. Keep the project: you need it in part D.

D) Make a print layout

Open the project of parts A–C. The result can look like this:

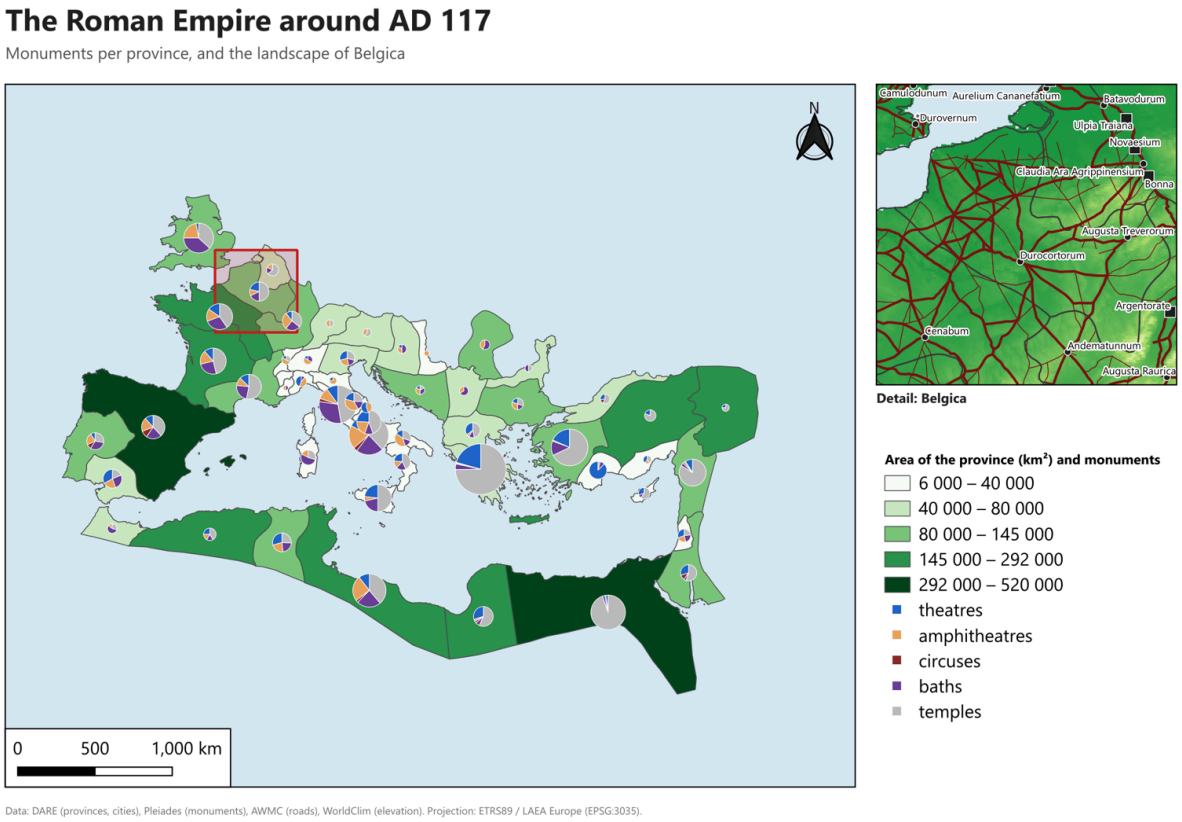
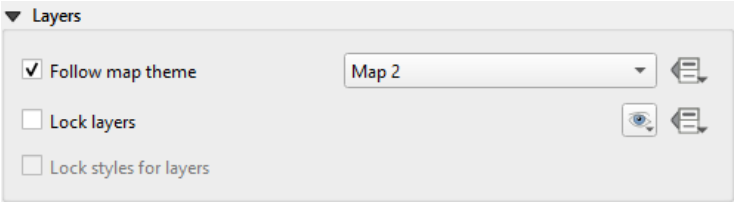


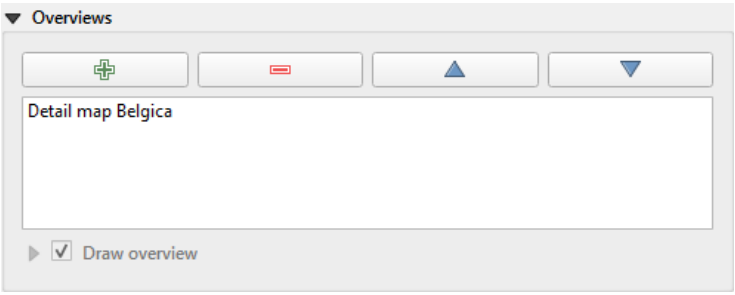
Figure 3: An example of the finished layout: the overview map follows the map theme “Map 2”, the detail map follows “Map 1”. The red frame shows the extent of the detail map.

1. **New layout.** Choose *Project* ▸ *New Print Layout…* and give the layout a name.

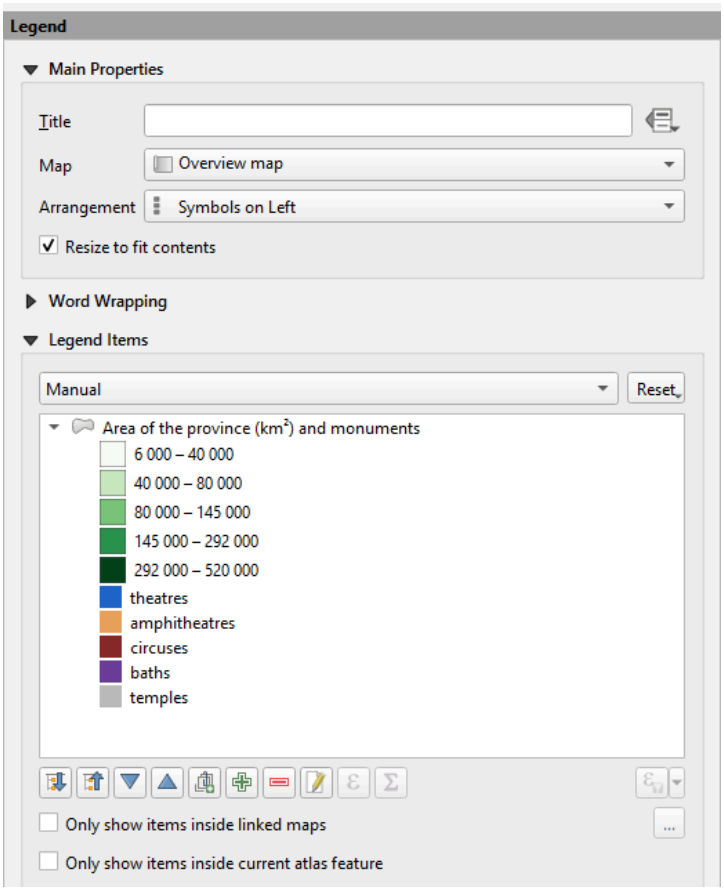
2. **Overview map.** Choose *Add Item* ▸ *Add Map* and drag a large rectangle on the page. In the *Item Properties* of this map, under *Layers*, tick *Follow map theme* and choose “Map 2”. Zoom the map to the whole empire.



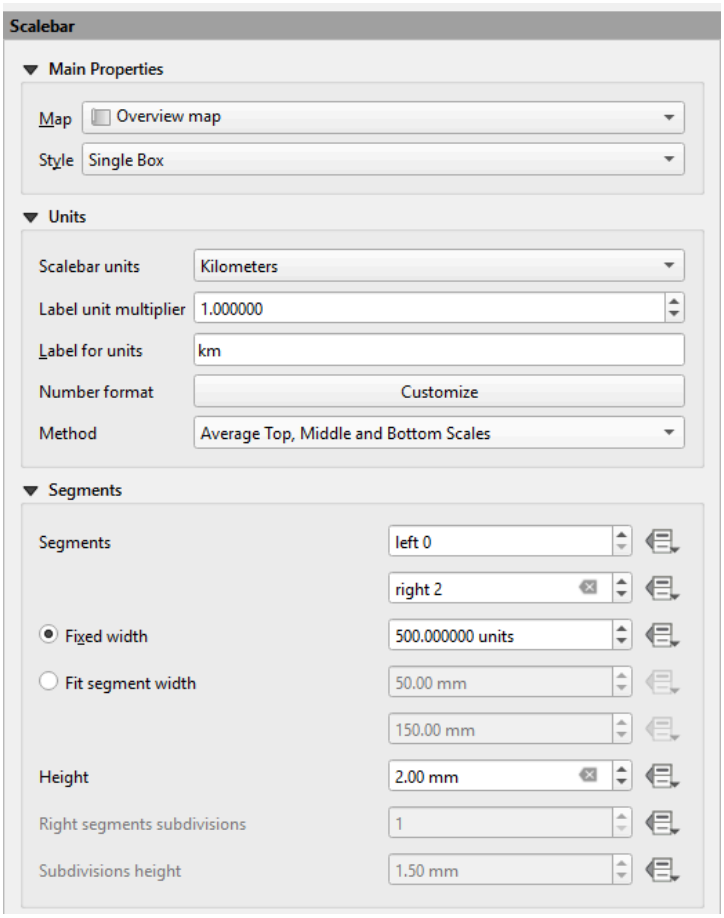
3. **Detail map.** Add a second, smaller map in the same way. Let it follow the map theme "Map 1". In the main QGIS window, zoom in on the province of Belgica. Back in the layout, click *Set Map Extent to Match Main Canvas Extent* in the *Item Properties* of the detail map.
4. **Extent frame.** Select the overview map. Under *Overviews*, click the plus button, and under *Map frame* choose the detail map. The overview map now shows a frame around the area of the detail map.



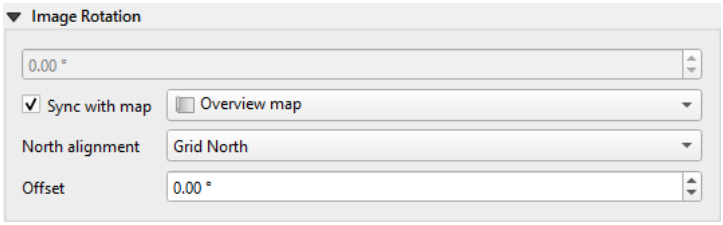
5. **Title, legend, scale bar and north arrow.**
   - Add a title with *Add Item* ▸ *Add Label*.
   - Add a legend with *Add Item* ▸ *Add Legend*. In its *Item Properties*, choose the overview map as the map it is linked to. Remove the legend items you do not need, or tick *Only show items inside linked maps*. To change the text of an item, double-click it.



- Add a scale bar with *Add Item* ▸ *Add Scale Bar* and link it to the overview map. If the numbers are long and overlap, set *Scalebar units* to *Kilometers*.



- Add a north arrow with *Add Item* ▸ *Add North Arrow*. Under *Image Rotation*, tick *Sync with map* and choose the overview map.



6. **Export.** Choose *Layout* ▸ *Export as PDF…*.